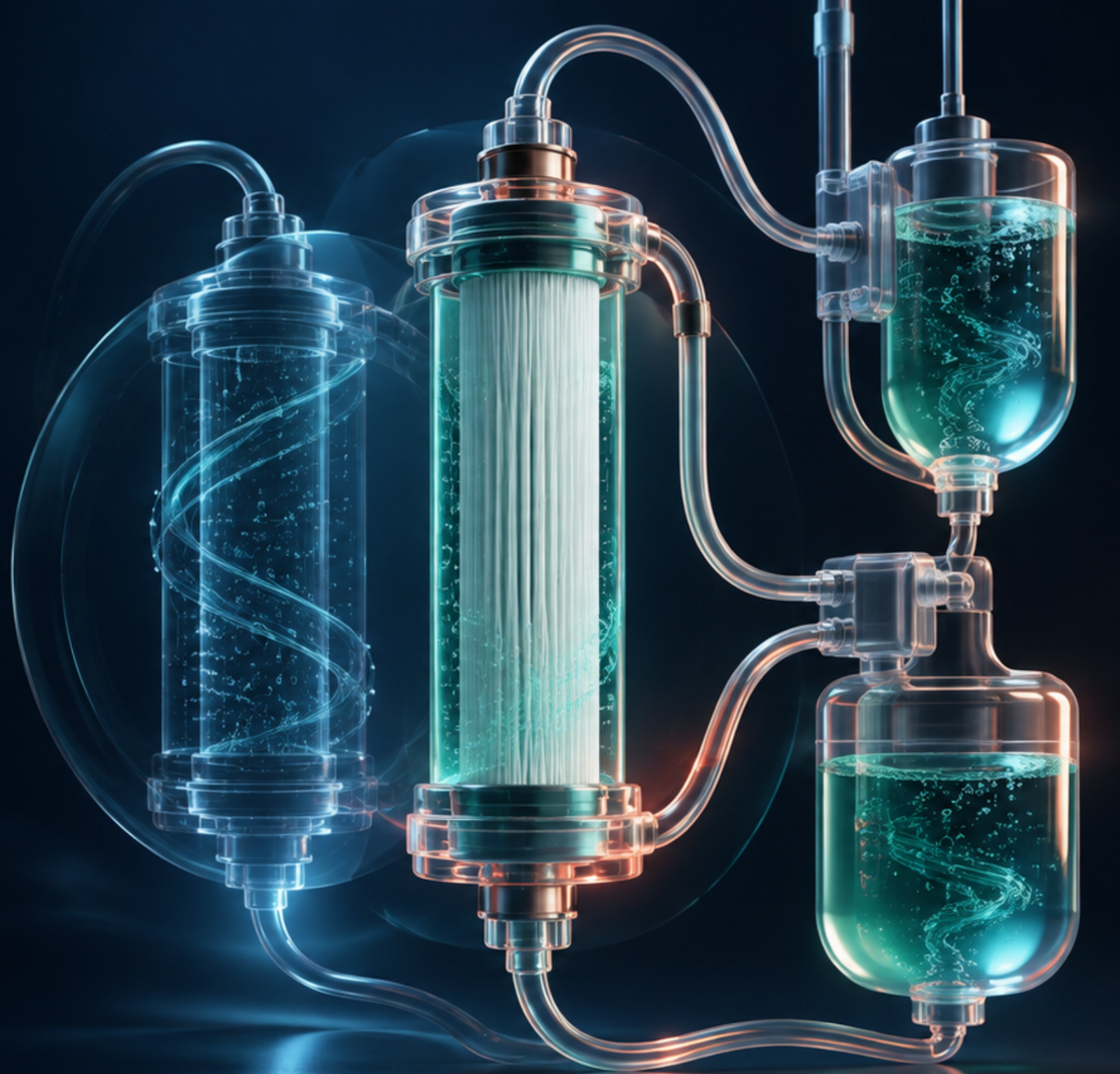
RENALIS

Evidence-aware dialysis digital twin

Competing explanations.
A better next measurement question.

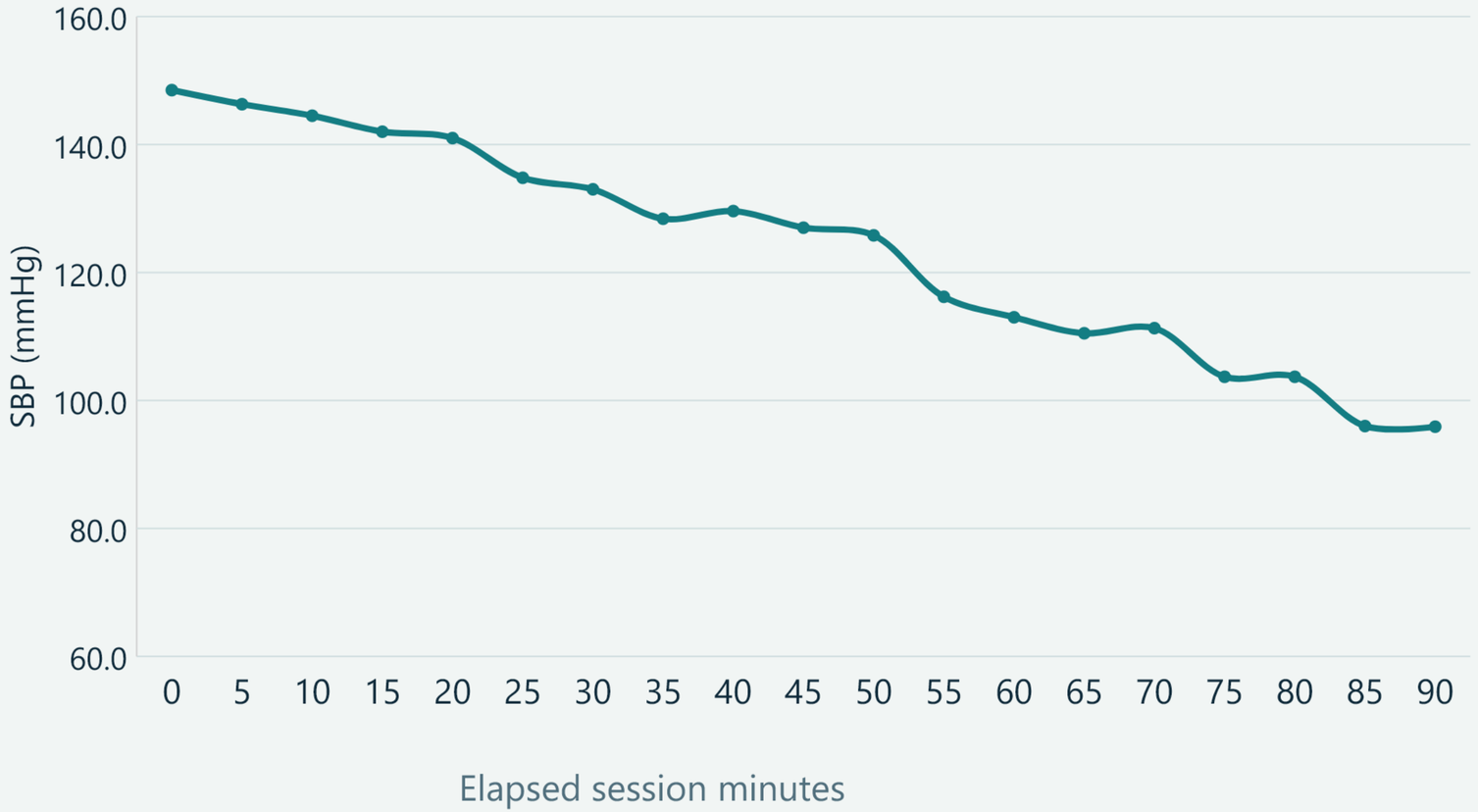


TEAM SIGMOID

Rehaan / JSS University, Noida

One BP trace can support several explanations

An observation can narrow uncertainty without identifying a unique physiological cause.



The missing variable

Is the pressure change more compatible with volume, vascular or cardiac response?

Synthetic replay at minute 90

The research workspace

One evidence cutoff connects the patient replay, hypotheses and acquisition history.

01 **Learn**

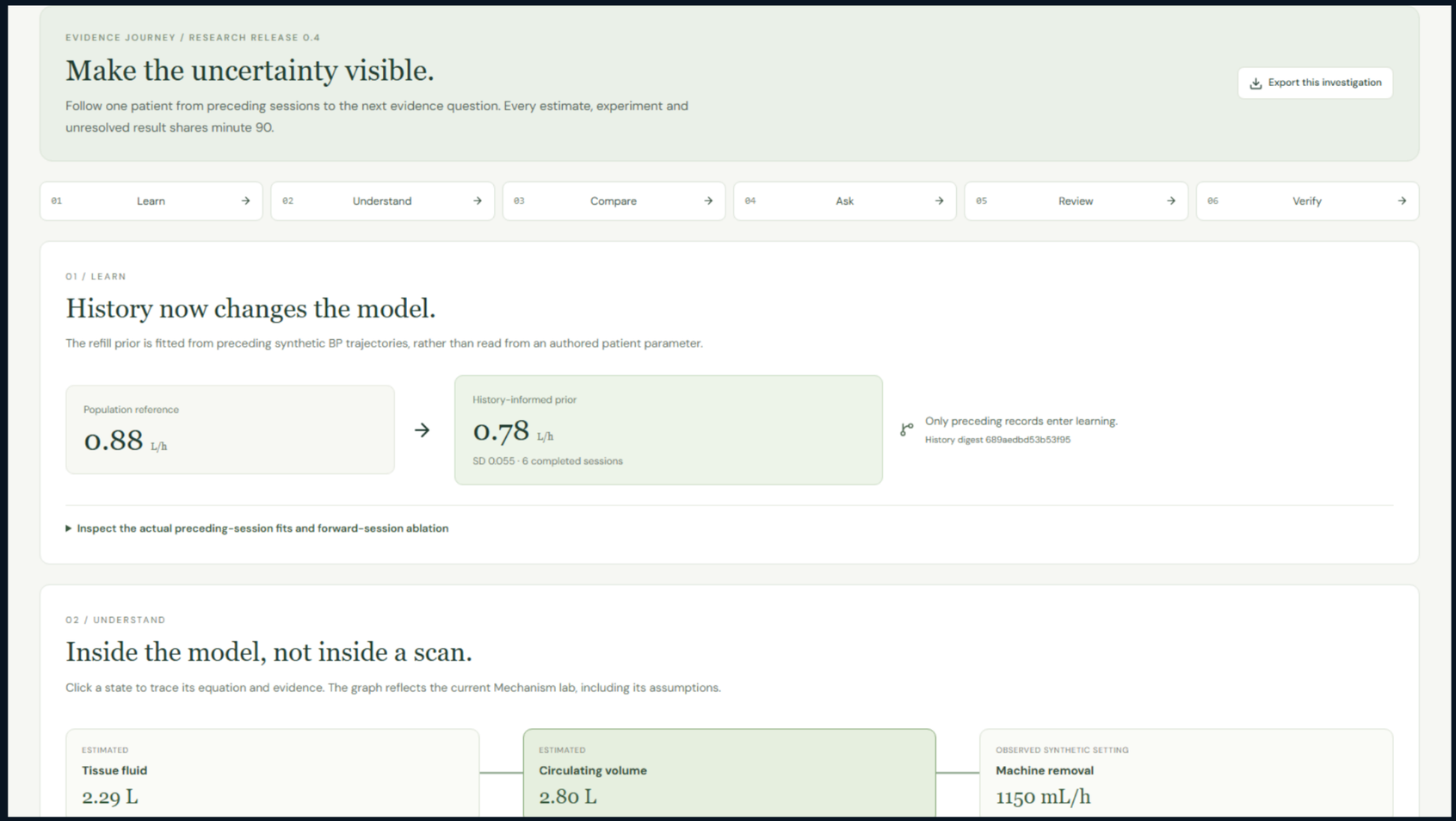
02 **Understand**

03 **Compare**

04 **Ask**

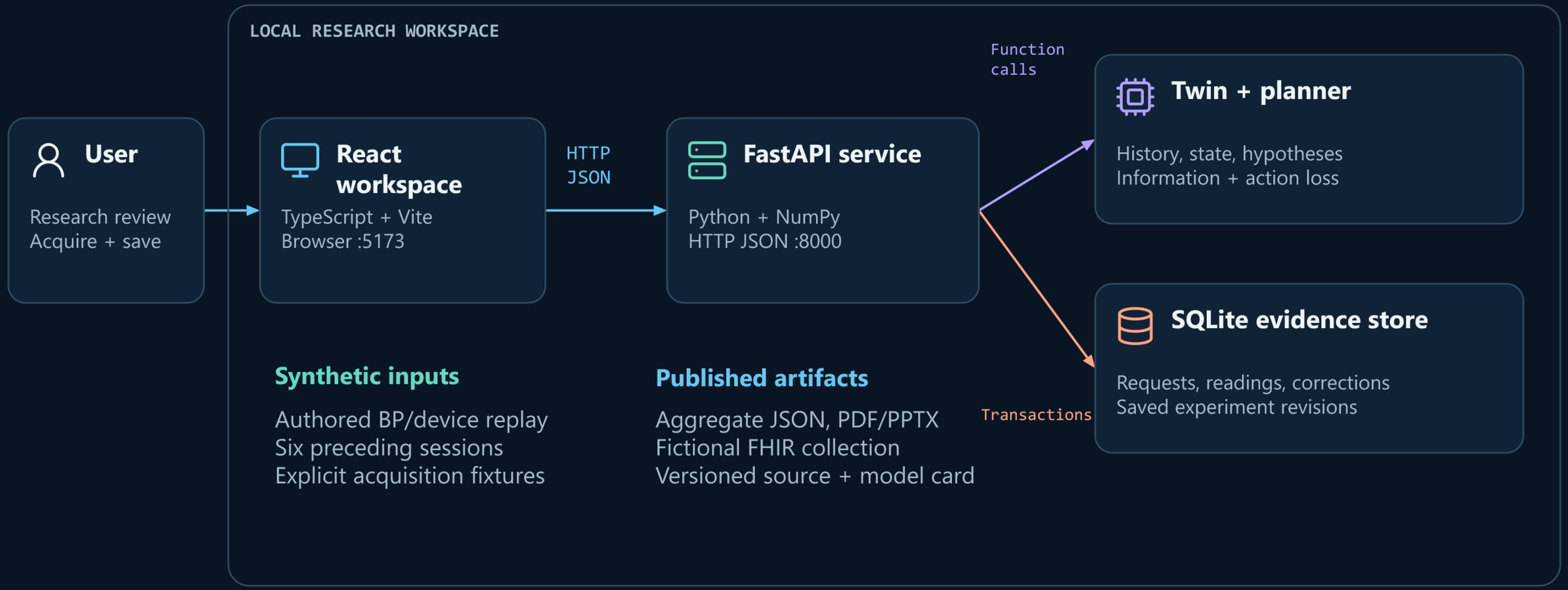
05 **Review**

06 **Verify**



System containers

Current local implementation. Browser requests stay separate from offline evaluation.



Observed HEMOBP modeling runs offline and supplies evaluation reports, not the synthetic twin state.

The prior comes from preceding sessions

Six completed synthetic sessions inform refill estimation before the current replay.



What the interface exposes

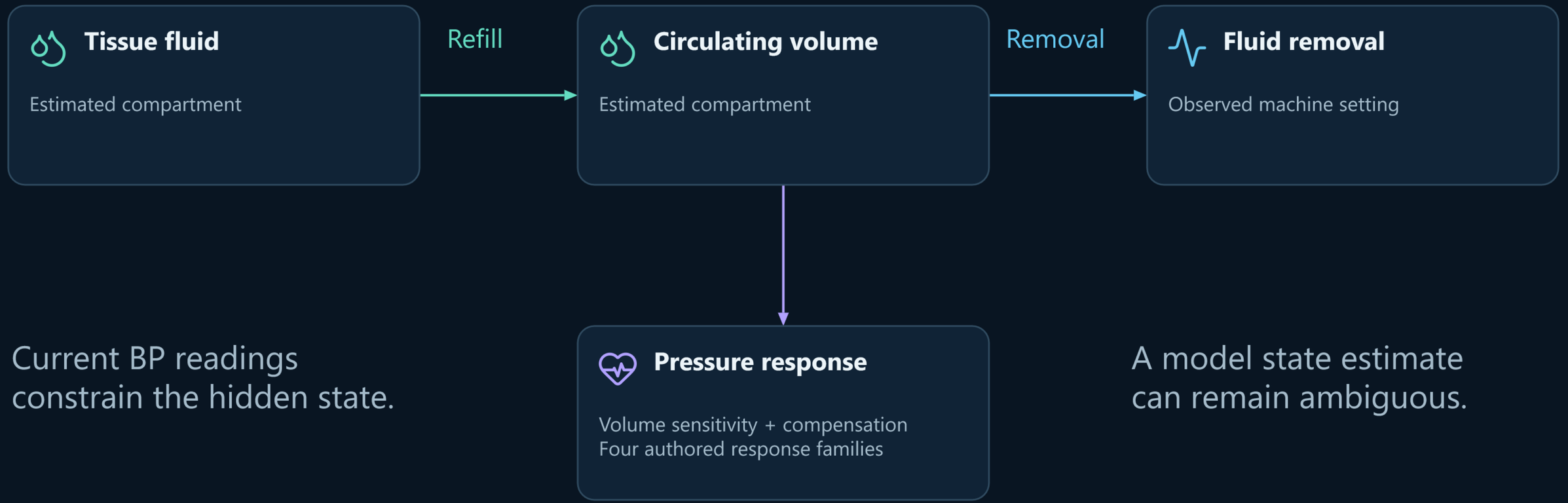
Per-session fits, the fitted prior and the forward-session ablation make the learner inspectable.

6

preceding completed sessions

The physiological model

Estimated compartments connect removal, refill and family-specific pressure response.



Current BP readings constrain the hidden state.

A model state estimate can remain ambiguous.

Four families fit the available past

Similar fit does not imply the same projected response.

Volume / refill

24.9%

relative model support

1.87 mmHg

past-fit RMSE

Vascular response

25.0%

relative model support

1.86 mmHg

past-fit RMSE

Cardiac response

25.0%

relative model support

1.83 mmHg

past-fit RMSE

Mixed response

25.1%

relative model support

1.80 mmHg

past-fit RMSE

Cardiac output narrows the disagreement

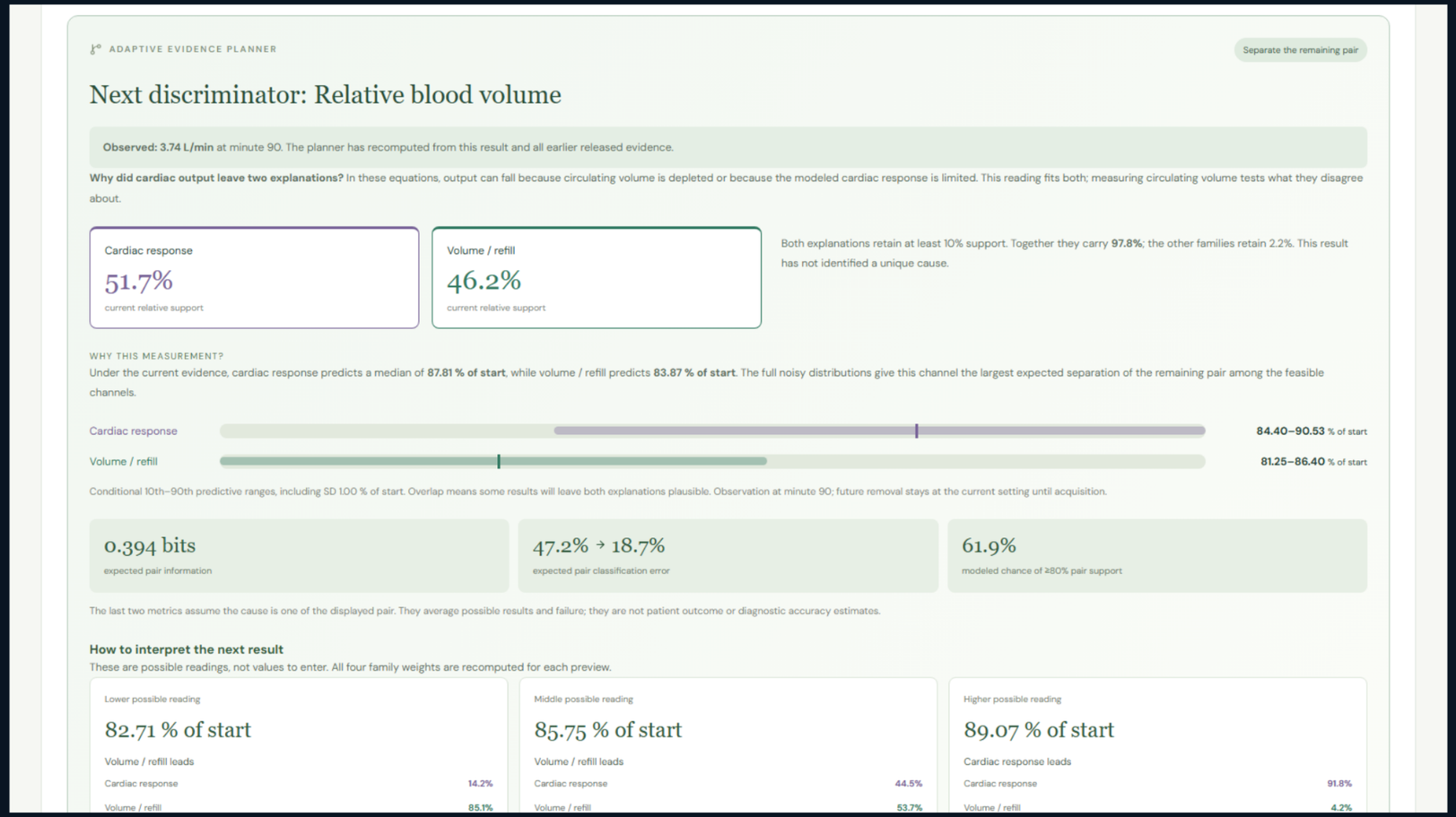
An acquired low-output result still fits both volume depletion and limited cardiac response.

3.74

L/min acquired cardiac output

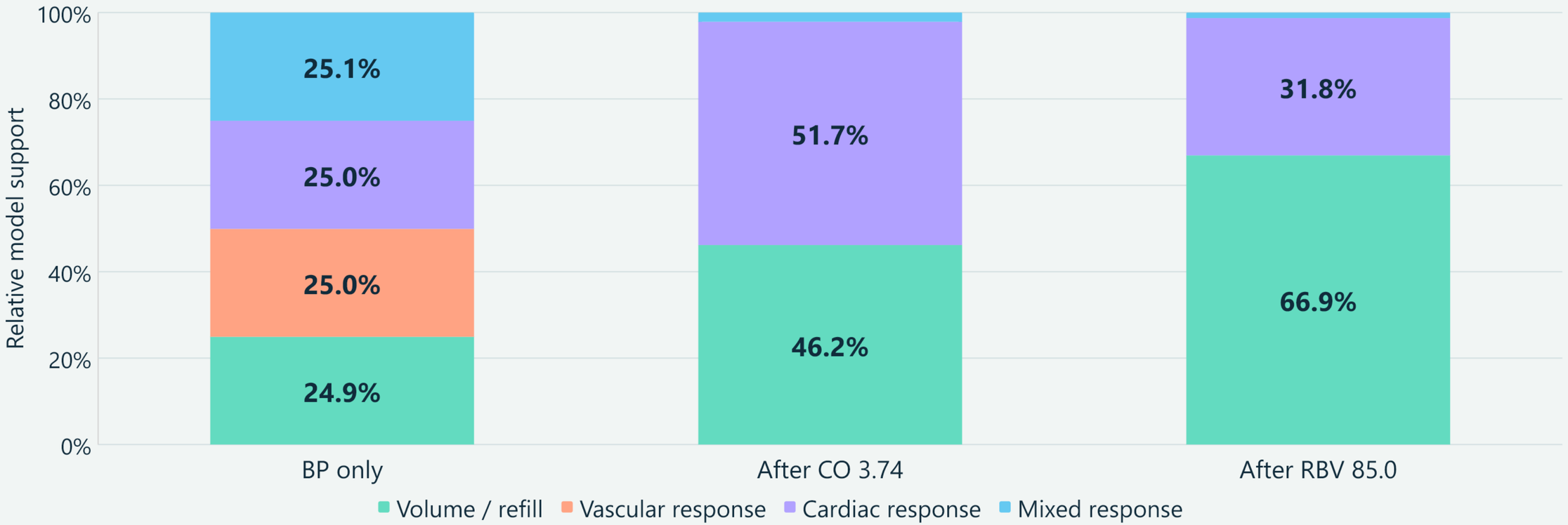
Cardiac 51.7%
Volume 46.2%

Current model support



How the acquired evidence changes support

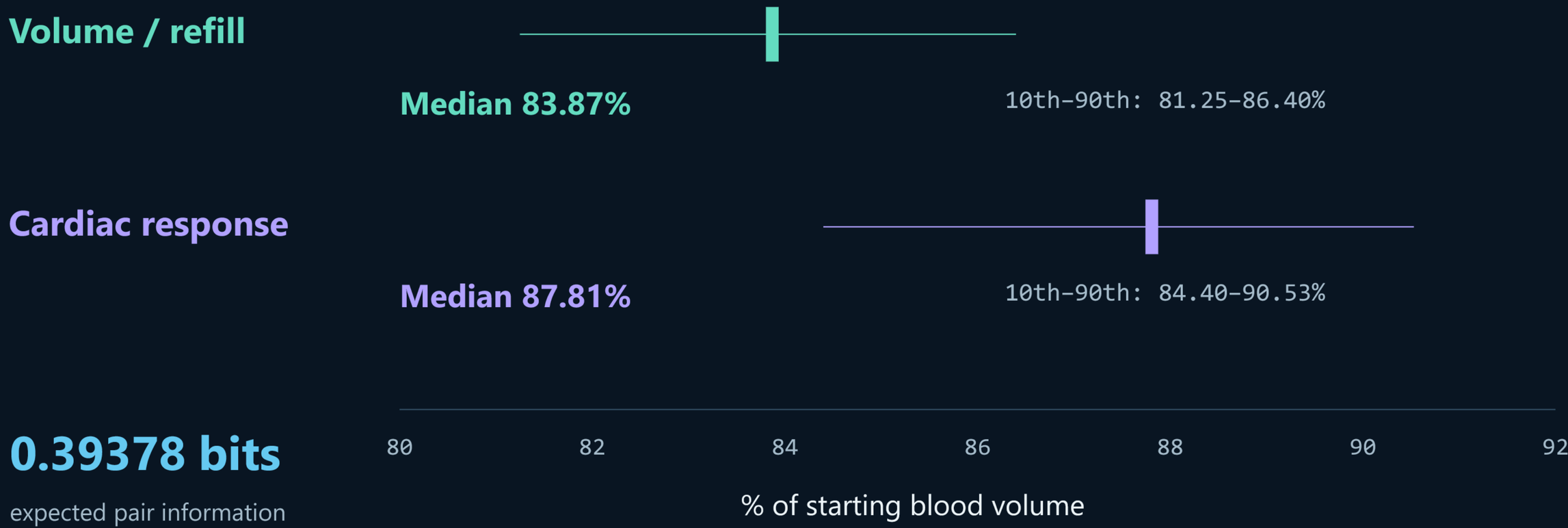
The planner recomputes the full bank after each result. Residual ambiguity remains visible.



Support is conditional on the authored model bank. It is not a probability of clinical diagnosis.

Why the next measurement is relative blood volume

The remaining families predict different noisy RBV distributions. Their ranges still overlap.

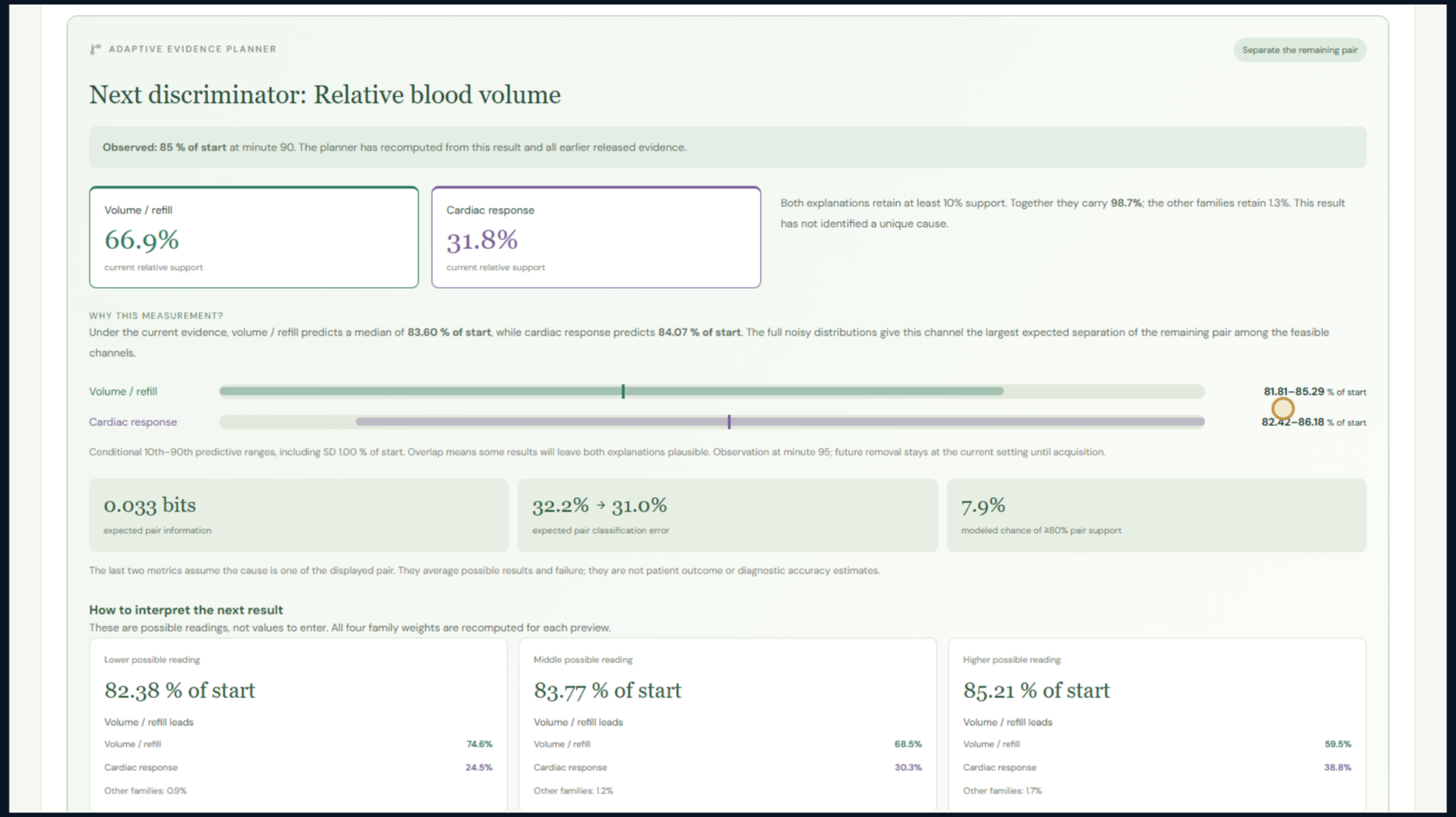


0.39378 bits

expected pair information

The device returns 85.0%

The acquired RBV shifts the balance toward volume / refill without eliminating cardiac support.



66.9%

volume / refill support

31.8%

cardiac response support

Information and action value answer different questions

Both use current beliefs, but only one includes the modeled consequence of a decision.



Information

Expected uncertainty reduction

Which feasible observation could separate the remaining explanations?

`Units: bits`



Action value

Expected modeled loss reduction

Could the result improve the comparison after waiting, failure and burden?

`Units: declared research loss`

Every observation has a lineage

Failures, artifacts and corrections remain distinguishable from evidence used in inference.

01 **Observe**

Record physiological time and quality.

02 **Release**

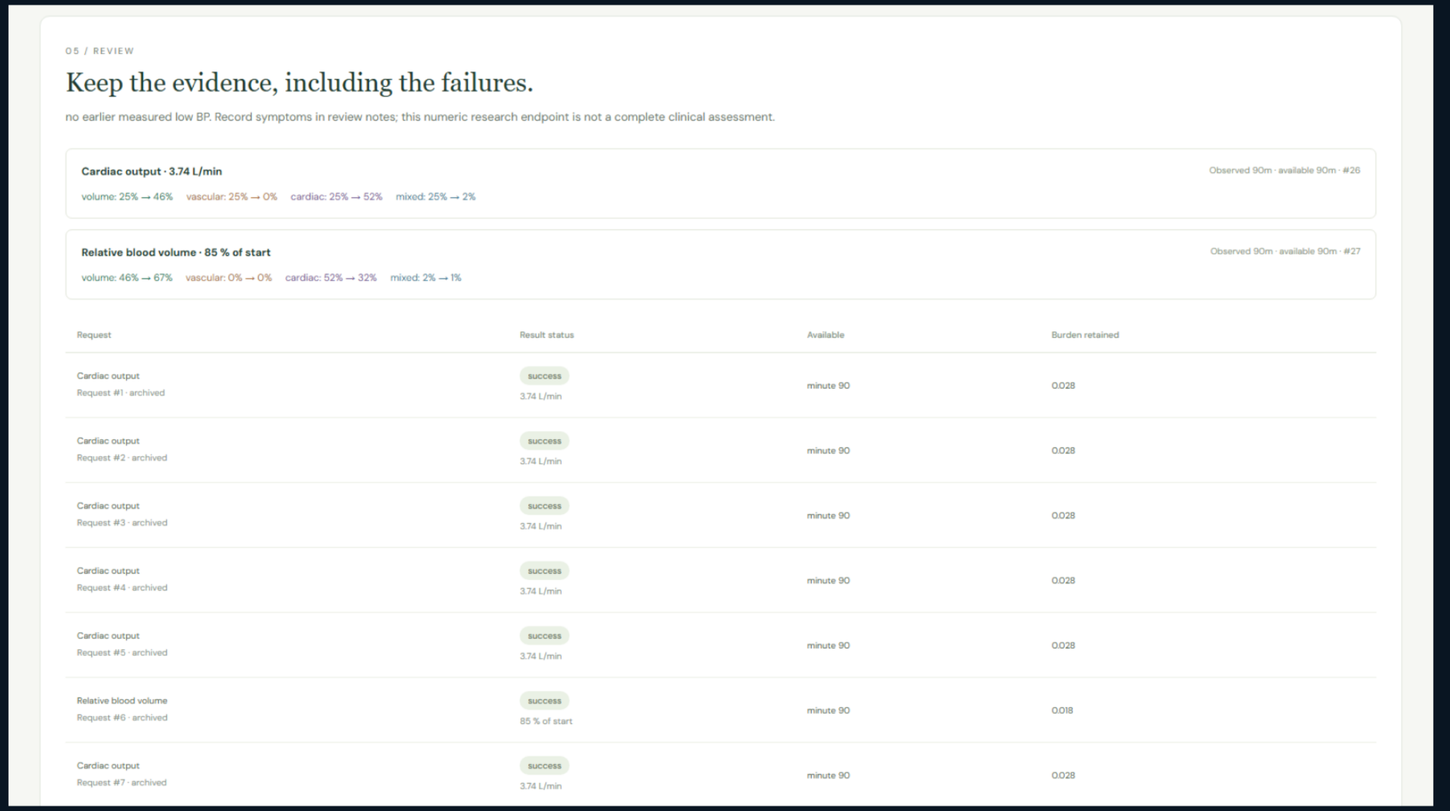
Make evidence available after delay.

03 **Correct**

Append a reason and supersedes link.

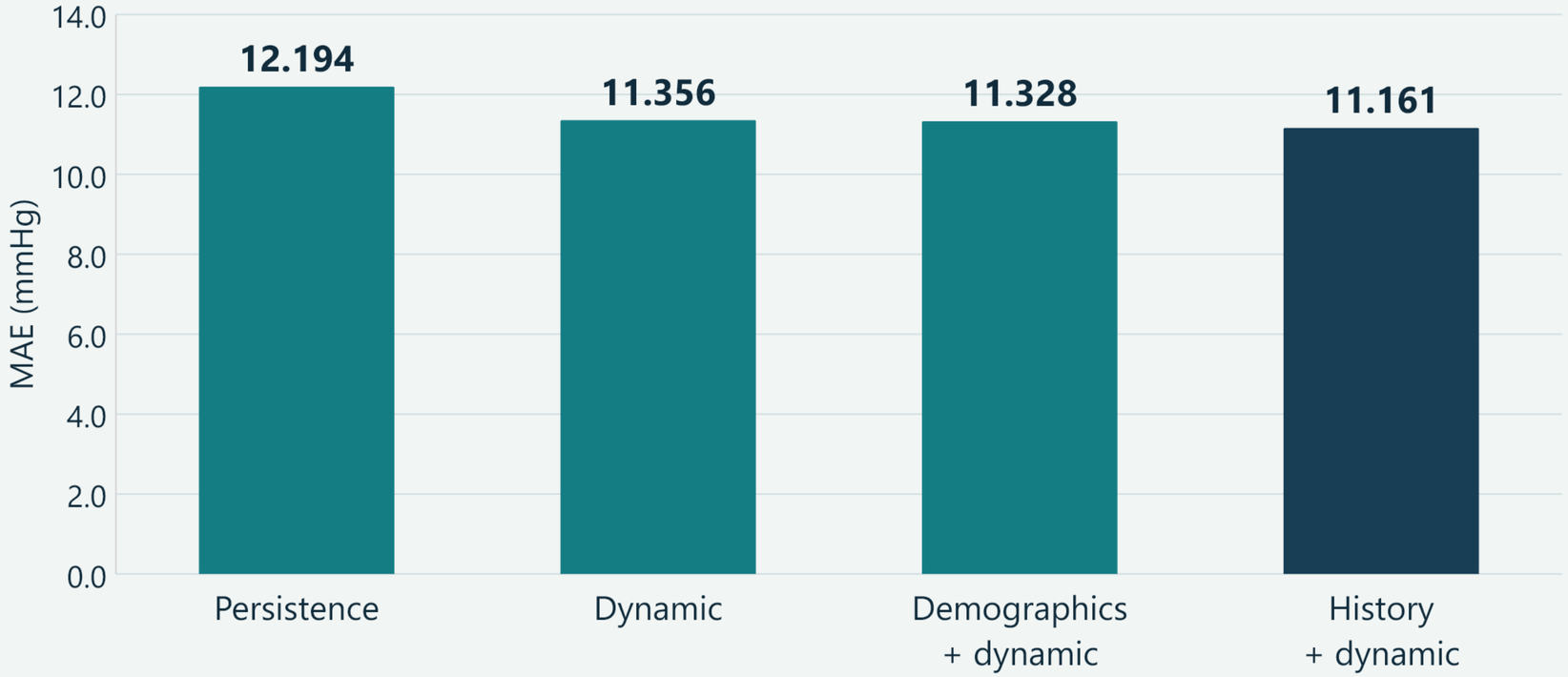
04 **Save**

Keep revision and evidence in a snapshot.



History adds a small retrospective signal

HEMOBP v3. 61,947 test windows from 176 patients, with no patient overlap between splits.



−0.1665

mmHg paired history increment

95% patient-cluster interval [−0.2298, −0.1099]

Clinical importance remains unestablished.

The evaluation still has material limits

A retrospective prediction result does not establish physiological identification or clinical benefit.

61.25%

event sensitivity

13.51%

positive predictive value

86.74%

coverage of nominal 90% range

Clock alignment

The empirical 8-hour shift needs independent confirmation.

Cohort exclusions

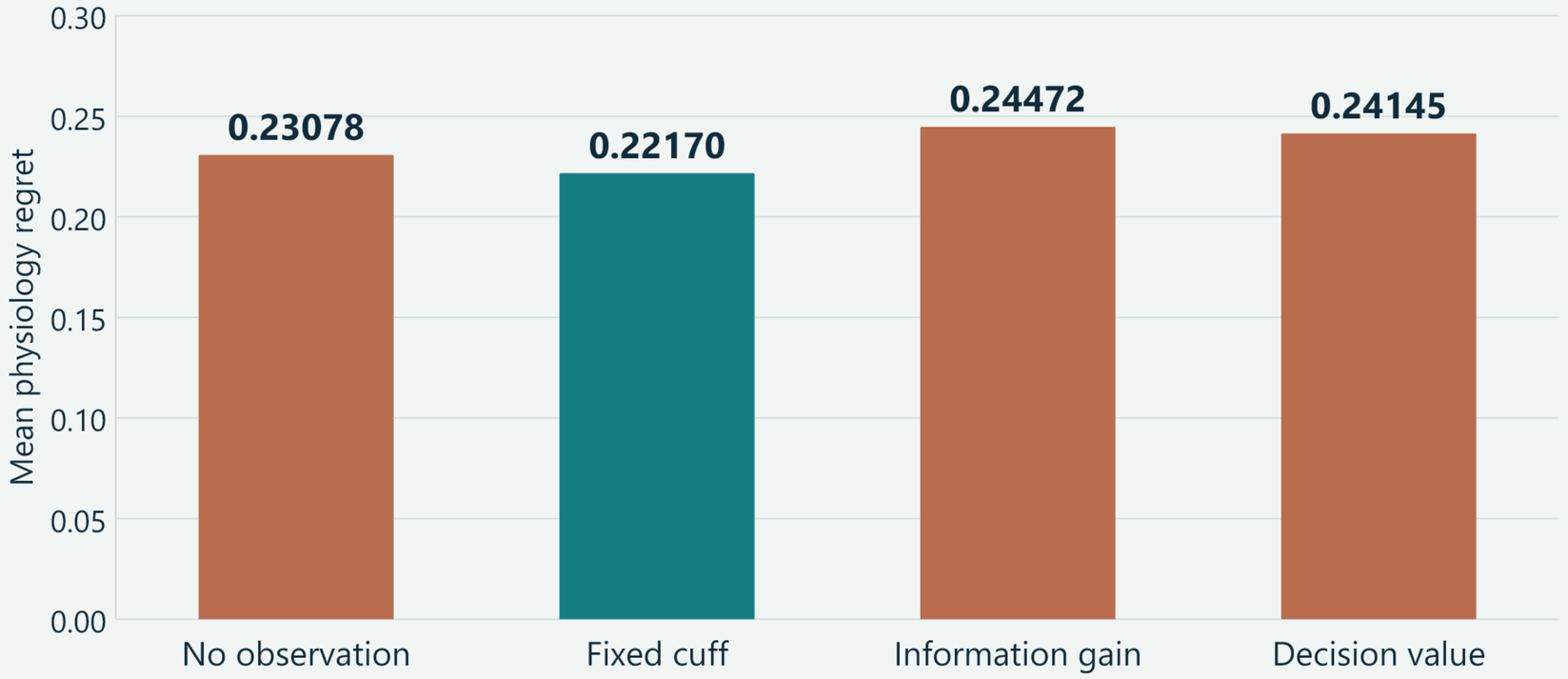
2,516,989 of 4,366,298 packets match the declared checks.

Endpoint scope

Next measured BP differs from continuous hypotension.

Decision value does not beat cuff overall

Independent synthetic holdout: 192 cases, four policies and one acquisition per policy.



0.22170

fixed cuff

0.24145

decision value

Lower is better.

Authority changes with evidence quality

The service withholds scenario authority when the replay cannot support it.



Current low BP

Current event is already present.
No advance-warning claim.



Stale or missing BP

Refresh the available evidence before comparing scenarios.



Completed session

Do not compare new removal profiles after session completion.



Contradictory evidence

Show poor fit and residual uncertainty in the model bank.

Extensions expose the next integration boundary

Current exports and transport experiments remain inspectable research components.

Current compartments

Evidence-weighted volume state

Longitudinal transport

Authored intake / output
Mass-balance checks

Synthetic FHIR R4

Patient / Device / BP Observation
Reference + unit checks

FUTURE EXTERNAL INTEGRATION



Clinical systems

Live EHR access, governance and ABDM conformance require independent work.

The next evidence gates

The strongest next work tests the assumptions that limit interpretation.

01

Source timing

Confirm HEMOBP clocks and reproduce the cohort.

02

Human interpretation

Test whether reviewers understand ambiguity.

03

Matched physiology

Calibrate the model against paired clinical measurements.

04

Independent policies

Predeclare multi-acquisition budgets and correlated noise.

05

External silent study

Measure calibration and workflow burden externally.

A clearer next evidence question

RENALIS makes the model disagreement, the next observation and the remaining uncertainty visible.

See the actual 20-minute demonstration

youtu.be/ZcWNYMtM1tI

github.com/RAK2315/RENALIS

Team Sigmoid

Rehaan
JSS University, Noida

Evidence and methods

Model card + source modules
Observed and policy reports
Research dossier + study protocol